

# Calculus II

## Homework

### Integration by parts

1. Let  $x \in (0, 1)$ . Express the following using  $x$  and  $\sqrt{1 - x^2}$ .

(a)  $\sin(\arcsin(x))$ .

(e)  $\sin(2 \arccos(x))$ .

(b)  $\sin(2 \arcsin(x))$ .

(f)  $\sin(3 \arccos(x))$ .

(c)  $\sin(3 \arcsin(x))$ .

(g)  $\cos(2 \arcsin(x))$ .

(d)  $\sin(\arccos(x))$ .

(h)  $\cos(3 \arccos(x))$ .

**Solution.** 1.b. Let  $y = \arcsin x$ . Then  $\sin y = x$ , and we can draw a right triangle with opposite side length  $x$  and hypotenuse length 1 to find the other trigonometric ratios of  $y$ .



Then  $\cos y = \frac{\sqrt{1-x^2}}{1} = \sqrt{1-x^2}$ . Now we use the double angle formula to find  $\sin(2 \arcsin x)$ .

$$\begin{aligned}\sin(2 \arcsin x) &= \sin(2y) \\ &= 2 \sin y \cos y \\ &= 2x\sqrt{1-x^2}.\end{aligned}$$

**Solution.** 1.c. Use the result of Problem 1.b. This also requires the addition formula for sine:

$$\sin(A + B) = \sin A \cos B + \sin B \cos A,$$

and the double angle formula for cosine:

$$\cos(2y) = \cos^2 y - \sin^2 y.$$

$$\begin{aligned}\sin(3 \arcsin x) &= \sin(3y) \\ &= \sin(2y + y) \\ &= \sin(2y) \cos y + \sin y \cos(2y) && \left| \begin{array}{l} \text{Use addition formula} \\ \text{Use double angle formulas} \end{array} \right. \\ &= (2 \sin y \cos y) \cos y + \sin y (\cos^2 y - \sin^2 y) \\ &= 2 \sin y \cos^2 y + \sin y \cos^2 y - \sin^3 y \\ &= 3 \sin y \cos^2 y - \sin^3 y \\ &= 3 \sin y (1 - \sin^2 y) - \sin^3 y \\ &= 3x(1 - x^2) - x^3 \\ &= 3x - 4x^3.\end{aligned}$$

The solution is complete. A careful look at the solution above reveals a strategy useful for problems similar to this one.

- Identify the inverse trigonometric expression-  $\arcsin x, \arccos x, \arctan x, \dots$ . In the present problem that was  $y = \arcsin x$ .
- The problem is therefore a trigonometric function of  $y$ .
- Using trig identities and algebra, rewrite the problem as a trigonometric expression involving only the trig function that transforms  $y$  to  $x$ . In the present problem we rewrote everything using  $\sin y$ .
- Use the fact that  $\sin(\arcsin x) = x, \cos(\arccos x) = x, \dots$ , etc. to simplify.

**Solution.** 1.f We use the same strategy outlined in the end of the solution of Problem 1.c. Set  $y = \arccos x$  and so  $\cos(y) = x$ . Therefore:

$$\begin{aligned}\sin(3y) &= \sin(2y + y) \\ &= \sin(2y) \cos y + \sin y \cos(2y) \\ &= 2 \sin y \cos y \cos y + \sin y (2 \cos^2 y - 1) \\ &= 2 \sin y \cos^2 y + \sin y (2 \cos^2 y - 1) \\ &= \sin y (4 \cos^2 y - 1) && \left| \begin{array}{l} \text{use } \cos y = x \\ \sin y = \sqrt{1-x^2} \end{array} \right. \\ &= \sqrt{1-x^2} (4x^2 - 1) .\end{aligned}$$

2. Express as the following as an algebraic expression of  $x$ . In other words, “get rid” of the trigonometric and inverse trigonometric expressions.

(a)  $\cos^2(\arctan x)$ .

(b)  $-\sin^2(\operatorname{arccot} x)$ .

(c)  $\frac{1}{\cos(\arcsin x)}$ .

$$\frac{x^2+1}{1}$$

$$\frac{x^2+1}{1}$$

(d)  $-\frac{1}{\sin(\arccos x)}$ .

$$\frac{x^2-1}{1}$$

$$\frac{x^2-1}{1}$$

**Solution.** 2.b. We follow the strategy outlined in the end of the solution of Problem 1.c. We set  $y = \operatorname{arccot} x$ . Then we need to express  $-\sin^2 y$  via  $\cot y$ . That is a matter of algebra:

$$\begin{aligned}
 -\sin^2(\operatorname{arccot} x) &= -\sin^2 y && \left| \begin{array}{l} \text{Set } y = \operatorname{arccot} x \\ \text{use } \sin^2 y + \cos^2 y = 1 \end{array} \right. \\
 &= -\frac{\sin^2 y}{\sin^2 y + \cos^2 y} \\
 &= -\frac{\sin^2 y}{1} \\
 &= -\frac{\sin^2 y + \cos^2 y}{\sin^2 y} \\
 &= -\frac{1}{1 + \cot^2 y} && \left| \begin{array}{l} \text{Substitute back } \cot y = x \end{array} \right. \\
 &= -\frac{1}{1 + x^2} .
 \end{aligned}$$

3. Rewrite as a rational function of  $t$ . This problem will be later used to derive the Euler substitutions (an important technique for integrating).

(a)  $\cos(2 \arctan t)$ .

(g)  $\cos(2 \operatorname{arccot} t)$ .

(b)  $\sin(2 \arctan t)$ .

(h)  $\sin(2 \operatorname{arccot} t)$ .

(c)  $\tan(2 \arctan t)$ .

(i)  $\tan(2 \operatorname{arccot} t)$ .

(d)  $\cot(2 \arctan t)$ .

(j)  $\cot(2 \operatorname{arccot} t)$ .

(e)  $\csc(2 \arctan t)$ .

(k)  $\csc(2 \operatorname{arccot} t)$ .

(f)  $\sec(2 \arctan t)$ .

(l)  $\sec(2 \operatorname{arccot} t)$ .

**Solution.** 3.a Set  $z = \arctan t$ , and so  $\tan z = t$ . Then

$$\begin{aligned}
 \cos(2 \arctan t) &= \cos(2z) \\
 &= \frac{\cos(2z)}{1} \\
 &= \frac{\cos^2 z - \sin^2 z}{\cos^2 z + \sin^2 z} \\
 &= \frac{(\cos^2 z - \sin^2 z) \frac{1}{\cos^2 z}}{(\sin^2 z + \cos^2 z) \frac{1}{\cos^2 z}} \\
 &= \frac{1 - \tan^2 z}{1 + \tan^2 z} \\
 &= \frac{1 - t^2}{1 + t^2} .
 \end{aligned}$$

use double angle formulas  
 and  $1 = \sin^2 z + \cos^2 z$   
 divide top and bottom by  $\cos^2 z$

**Solution.** 3.d Set  $z = \arctan t$ , and so  $\tan z = t$ . Then

$$\begin{aligned}
 \cot(2 \arctan t) &= \cot(2z) \\
 &= \frac{\cos(2z)}{\sin(2z)} \\
 &= \frac{\cos^2 z - \sin^2 z}{2 \sin z \cos z} \\
 &= \frac{1 - \tan^2 z}{2 \tan z} \\
 &= \frac{1 - t^2}{2t} \\
 &= \frac{1}{2} \left( \frac{1}{t} - t \right) .
 \end{aligned}$$

use double angle formulas

4. Compute the derivative (derive the formula).

(a)  $(\arctan x)'$ .

$$\frac{1}{1+x^2}$$

(d)  $(\arccos x)'$ .

$$-\frac{1}{\sqrt{1-x^2}}$$

(b)  $(\operatorname{arccot} x)'$ .

$$-\frac{1}{1+x^2}$$

(e) Let  $\operatorname{arcsec}$  denote the inverse of the secant function. Compute  $(\operatorname{arcsec} x)'$ .

$$\frac{1}{1-x^2} \frac{1}{\sqrt{1-x^2}}$$

(c)  $(\arcsin x)'$ .

$$\frac{1}{\sqrt{1-x^2}}$$

5. (a) Let  $a + b \neq k\pi$ ,  $a \neq k\pi + \frac{\pi}{2}$  and  $b \neq k\pi + \frac{\pi}{2}$  for any  $k \in \mathbb{Z}$  (integers). Prove that

$$\frac{\tan a + \tan b}{1 - \tan a \tan b} = \tan(a + b) \quad .$$

(b) Let  $x$  and  $y$  be real. Prove that, for  $xy \neq 1$ , we have

$$\arctan x + \arctan y = \arctan \left( \frac{x + y}{1 - xy} \right)$$

if the left hand side lies between  $(-\frac{\pi}{2}, \frac{\pi}{2})$ .

**Solution.** 5.a We start by recalling the formulas

$$\begin{aligned} \cos(a + b) &= \cos a \cos b - \sin a \sin b \\ \sin(a + b) &= \sin a \cos b + \sin b \cos a \quad . \end{aligned}$$

These formulas have been previously studied; alternatively they follow from Euler's formula and the computation

$$\begin{aligned} \cos(a + b) + i \sin(a + b) &= e^{i(a+b)} = e^{ia} e^{ib} = (\cos a + i \sin a)(\cos b + i \sin b) \\ &= \cos a \cos b - \sin a \sin b + i(\sin a \cos b + \sin b \cos a) \quad . \end{aligned}$$

Now 5.a is done via a straightforward computation:

$$\begin{aligned} \tan(a + b) &= \frac{\sin(a + b)}{\cos(a + b)} = \frac{\sin a \cos b + \sin b \cos a}{\cos a \cos b - \sin a \sin b} = \frac{(\sin a \cos b + \sin b \cos a) \frac{1}{\cos a \cos b}}{(\cos a \cos b - \sin a \sin b) \frac{1}{\cos a \cos b}} \\ &= \frac{\tan a + \tan b}{1 - \tan a \tan b} \quad . \end{aligned} \tag{1}$$

5.b is a consequence of 5.a. Let  $a = \arctan x$ ,  $b = \arctan y$ . Then (??) becomes

$$\tan(\arctan x + \arctan y) = \frac{\tan(\arctan x) + \tan(\arctan y)}{1 - \tan(\arctan x) \tan(\arctan y)} = \frac{x + y}{1 - xy} \quad ,$$

where we use the fact that  $\tan(\arctan w) = w$  for all  $w$ . We recall that  $\arctan(\tan z) = z$  whenever  $z \in (-\frac{\pi}{2}, \frac{\pi}{2})$ . Now take  $\arctan$  on both sides of the above equality to obtain

$$\arctan x + \arctan y = \arctan \left( \frac{x + y}{1 - xy} \right) \quad .$$

6. Evaluate the indefinite integral. Illustrate the steps of your solutions.

(a)  $\int x \sin x dx.$

$$-x \cos x + \sin x + C$$

(e)  $\int x^2 \cos(3x) dx.$

$$\frac{x^2}{2} \sin(3x) - \frac{x}{6} \cos(3x) + \frac{1}{36} \sin(3x) + C$$

(b)  $\int x e^{-x} dx.$

$$-x e^{-x} - e^{-x} + C$$

(f)  $\int x^2 e^{-2x} dx.$

$$-\frac{x^2}{2} e^{-2x} - \frac{x}{2} e^{-2x} - \frac{1}{4} e^{-2x} + C$$

(c)  $\int x^2 e^x dx.$

$$x^2 e^x - 2x e^x + 2e^x + C$$

(g)  $\int x \sin(2x) dx.$

$$-\frac{x}{2} \cos(2x) + \frac{1}{4} \sin(2x) + C$$

(d)  $\int x \sin(-2x) dx.$

$$-\frac{x}{2} \cos(-2x) - \frac{1}{4} \sin(-2x) + C$$

(h)  $\int x \cos(3x) dx.$

$$(i) \int x^2 e^{2x} dx. \quad \text{ANSWER: } \frac{1}{8} \cos(3x) + (x) \sin(3x) + \frac{5}{24} \sin(3x) \quad (j) \int x^3 e^x dx. \quad \text{ANSWER: } x^3 e^x - x^2 e^x + x e^x - e^x + C$$

$$\text{ANSWER: } \frac{1}{x^2} + \frac{2}{x^3} + x^2 e^{\frac{2}{x}} - x^2 e^{\frac{2}{x}} \frac{2}{x^2} \quad \text{ANSWER: } \frac{1}{2} \sin(2x) + x \cos(2x) + \frac{5}{8} \sin(2x)$$

**Solution. 6.a.**

$$\int x \underbrace{\sin x dx}_{=d(-\cos x)} = - \int x d(\cos x) = -x \cos x + \int \cos x dx = -x \cos x + \sin x + C.$$

**Solution. 6.c.**

$$\begin{aligned} \int x^2 \underbrace{e^x dx}_{d(e^x)} &= \int x^2 de^x = x^2 e^x - \int e^x 2x dx = x^2 e^x - \int 2x de^x \\ &= x^2 e^x - 2x e^x + \int 2e^x dx = x^2 e^x - 2x e^x + 2e^x + C. \end{aligned}$$

**Solution. 6.f.**

$$\begin{aligned} \int x^2 e^{-2x} dx &= \int x^2 d\left(\frac{e^{-2x}}{-2}\right) && \left| \text{Integrate by parts} \right. \\ &= -\frac{x^2 e^{-2x}}{2} - \int \left(\frac{e^{-2x}}{-2}\right) d(x^2) \\ &= -\frac{x^2 e^{-2x}}{2} + \int x e^{-2x} dx \\ &= -\frac{x^2 e^{-2x}}{2} + \int x d\left(\frac{e^{-2x}}{-2}\right) && \left| \text{Integrate by parts} \right. \\ &= -\frac{x^2 e^{-2x}}{2} - \frac{x e^{-2x}}{2} + \frac{1}{2} \int e^{-2x} dx \\ &= -\frac{x^2 e^{-2x}}{2} - \frac{x e^{-2x}}{2} - \frac{e^{-2x}}{4} + C. \end{aligned}$$

7. Evaluate the indefinite integral. Illustrate the steps of your solutions.

$$(a) \int x^2 \cos(2x) dx. \quad \text{ANSWER: } \frac{1}{8} \cos(2x) + x \sin(2x) + \frac{5}{24} \sin(2x) \quad (h) \int e^{-\sqrt{x}} dx. \quad \text{ANSWER: } -2\sqrt{x} e^{-\sqrt{x}} - 2e^{-\sqrt{x}} + C$$

$$(b) \int x^2 e^{ax} dx, \text{ where } a \text{ is a constant.} \quad \text{ANSWER: } \frac{1}{2} \sin(2x) + x \cos(2x) + \frac{5}{8} \sin(2x) \quad (i) \int \cos^2 x dx. \quad \text{ANSWER: } \frac{1}{2} \sin(2x) + \frac{1}{2} x$$

$$(c) \int x^2 e^{-ax} dx, \text{ where } a \text{ is a constant.} \quad \text{ANSWER: } \frac{1}{2} \sin(2x) + x \cos(2x) + \frac{5}{8} \sin(2x) \quad (j) \int \frac{x}{1+x^2} dx \text{ (Hint: use substitution rule, don't use integration by parts)} \quad \text{ANSWER: } \frac{1}{2} \ln(1+x^2) + C$$

$$(d) \int x^2 \frac{(e^{ax} + e^{-ax})^2}{4} dx, \text{ where } a \text{ is a constant.} \quad \text{ANSWER: } \frac{1}{2} \sin(2x) + x \cos(2x) + \frac{5}{8} \sin(2x) \quad (k) \int (\arctan x) dx. \quad \text{ANSWER: } x \arctan x - \frac{1}{2} \ln(1+x^2) + C$$

$$(e) \int \frac{1}{\cos^2 x} dx. \quad \text{(Hint: This problem does not require integration by parts. What is the derivative of } \tan x?) \quad \text{ANSWER: } \frac{1}{2} \sin(2x) + x \cos(2x) + \frac{5}{8} \sin(2x) \quad (l) \int (\arcsin x) dx. \quad \text{ANSWER: } x \arcsin x + \sqrt{1-x^2} + C$$

$$(f) \int (\tan^2 x) dx. \quad \text{(Hint: This problem does not require integration by parts. We can use } \tan^2 x = \frac{1}{\cos^2 x} - 1 \text{ and the previous problem.)} \quad (m) \int (\arcsin x)^2 dx. \quad \text{(Hint: Try substituting } x = \sin y.) \quad \text{ANSWER: } x \arcsin x + \sqrt{1-x^2} - 2 + C$$

$$(g) \int x \tan^2 x dx. \quad \text{(Hint: } \tan^2 x dx = d(F(x)), \text{ where } F(x) \text{ is the answer from the preceding problem.)} \quad (n) \int \arctan\left(\frac{1}{x}\right) dx.$$

$$(o) \int \sin x e^x dx$$

$$(p) \int \cos x e^x dx$$

$$(q) \int \sin(\ln(x)) dx.$$

$$(r) \int \cos(\ln(x)) dx.$$

$$(s) \int \ln x dx$$

$$(t) \int x \ln x dx.$$

$$C + (x \sec x e^x - x \sin x e^x) \frac{e^x}{2} \quad \text{ANSWER}$$

$$(u) \int \frac{\ln x}{\sqrt{x}} dx.$$

$$C + (x \ln x e^x + x \cos x e^x) \frac{e^x}{2} \quad \text{ANSWER}$$

$$(v) \int (\ln x)^2 dx.$$

$$C + ((x \ln x) \cos x - (x \ln x) \sin x) \frac{e^x}{2} \quad \text{ANSWER}$$

$$(w) \int (\ln x)^3 dx.$$

$$C + ((x \ln x) \sin x + (x \ln x) \cos x) \frac{e^x}{2} \quad \text{ANSWER}$$

$$(x) \int x^2 \cos^2 x dx. \quad (\text{This problem is related to Problem 7.d})$$

$$\text{as } \cos x = \frac{e^{ix} + e^{-ix}}{2}.$$

$$e^{ix} \frac{9}{4} + (x^2) \ln \frac{8}{4} - (x^2) \cos x \frac{5}{4} + (x^2) \sin x \frac{3}{4} \quad \text{ANSWER}$$

**Solution. 7.g.**

$$\begin{aligned} \int x \tan^2 x dx &= \int x (\sec^2 x - 1) dx && \left| \text{use } \sec^2 x - 1 = \tan^2 x \right. \\ &= \int x (\sec^2 x - 1) dx \\ &= -\int x dx + \int x \sec^2 x dx && \left| \text{use } d(\tan x) = \sec^2 x dx \right. \\ &= -\frac{x^2}{2} + \int x d(\tan x) && \left| \text{integrate by parts} \right. \\ &= -\frac{x^2}{2} + x \tan x - \int \tan x dx \\ &= -\frac{x^2}{2} + x \tan x - \int \frac{\sin x}{\cos x} dx && \left| \text{use } \sin x dx = -d(\cos x) \right. \\ &= -\frac{x^2}{2} + x \tan x + \int \frac{d(\cos x)}{\cos x} && \left| \text{Set } y = \cos x \right. \\ &= -\frac{x^2}{2} + x \tan x + \int \frac{1}{y} dy \\ &= -\frac{x^2}{2} + x \tan x + \ln |y| + C && \left| \text{Substitute back } y = \cos x \right. \\ &= -\frac{x^2}{2} + x \tan x + \ln |\cos x| + C \quad . \end{aligned}$$

**Solution. 7.h.**

$$\begin{aligned} \int e^{-\sqrt{x}} dx &= \int 2ye^{-y} dy && \left| \begin{array}{l} \text{Subst.: } \sqrt{x} = y \\ \frac{1}{2\sqrt{x}} dx = dy \\ dx = 2y dy \end{array} \right. \\ &= \int 2y d(-e^{-y}) && \left| \text{int. by parts} \right. \\ &= -2ye^{-y} + 2 \int e^{-y} dy \\ &= -2ye^{-y} - 2e^{-y} + C \\ &= -2\sqrt{x}e^{-\sqrt{x}} - 2e^{-\sqrt{x}} + C \quad . \end{aligned}$$

**Solution. 7.i.** Later, we shall study general methods for solving trigonometric integrals that will cover this example. Let us however show one way to solve this integral by integration by parts.

$$\begin{aligned}
\int \cos^2 x dx &= x \cos^2 x - \int x d(\cos^2 x) \\
&= x \cos^2 x - \int x 2 \cos x (-\sin x) dx && \left| \sin(2x) = 2 \sin x \cos x \right. \\
&= x \cos^2 x + \int x \sin(2x) dx \\
&= x \cos^2 x + \int x d\left(\frac{-\cos(2x)}{2}\right) \\
&= x \cos^2 x + x \left(\frac{-\cos(2x)}{2}\right) - \int \left(\frac{-\cos(2x)}{2}\right) dx \\
&= \frac{x}{2} (2 \cos^2 x - \cos(2x)) + \frac{\sin(2x)}{4} + C && \left| \cos(2x) = \cos^2 x - \sin^2 x \right. \\
&= \frac{x}{2} (2 \cos^2 x - (\cos^2 x - \sin^2 x)) + \frac{\sin(2x)}{4} + C && \left| \cos^2 x + \sin^2 x = 1 \right. \\
&= \frac{x}{2} + \frac{\sin(2x)}{4} + C .
\end{aligned}$$

**Solution. 7.k**

$$\begin{aligned}
\int \arctan x dx &= x \arctan x - \int x d(\arctan x) \\
&= x \arctan x - \int \frac{x}{x^2 + 1} dx \\
&= x \arctan x - \int \frac{\frac{1}{2} d(x^2)}{x^2 + 1} \\
&= x \arctan x - \int \frac{\frac{1}{2} d(x^2 + 1)}{x^2 + 1} \\
&= x \arctan x - \frac{1}{2} \ln(x^2 + 1) + C .
\end{aligned}$$

**Solution. 7.m.**

$$\begin{aligned}
\int (\arcsin x)^2 dx &= \int (\arcsin(\sin y))^2 d(\sin y) && \left| \text{Set } x = \sin y \right. \\
&= \int y^2 \cos y dy = \int y^2 d(\sin y) && \left| \text{Integrate by parts} \right. \\
&= y^2 \sin y - \int 2y \sin y dy \\
&= y^2 \sin y + \int 2y d(\cos y) && \left| \text{Integrate by parts} \right. \\
&= y^2 \sin y + 2y \cos y - 2 \int \cos y dy \\
&= y^2 \sin y + 2y \cos y - 2 \sin y + C && \left| \text{Substitute } y = \arcsin x \right. \\
&= x(\arcsin x)^2 \\
&\quad + 2\sqrt{1 - x^2} \arcsin x - 2x + C .
\end{aligned}$$

**Solution. 7.o**

$$\begin{aligned}
\int \sin x \underbrace{e^x dx}_{=de^x} &= \sin x e^x - \int e^x d(\sin x) = \sin x e^x - \int \cos x \underbrace{e^x dx}_{=de^x} \\
&= \sin x e^x - e^x \cos x + \int e^x d(\cos x) \\
&= e^x \sin x - e^x \cos x - \int e^x \sin x dx && \left| \text{add } \int e^x \sin x dx \text{ to both sides} \right. \\
2 \int \sin x e^x dx &= \sin x e^x - e^x \cos x \\
\int \sin x e^x dx &= \frac{1}{2} (\sin x e^x - e^x \cos x) .
\end{aligned}$$

**Solution. 7.q.**



$$\begin{aligned}
\int \sin(\ln x) dx &= x \sin(\ln x) - \int x d(\sin(\ln x)) && \left| \text{int. by parts} \right. \\
&= x \sin(\ln x) - \int x (\cos(\ln x)) (\ln x)' dx \\
&= x \sin(\ln x) - \int \cos(\ln x) dx && \left| \text{int. by parts} \right. \\
&= x \sin(\ln x) - \left( x \cos(\ln x) - \int x d(\cos(\ln x)) \right) \\
&= x \sin(\ln x) - x \cos(\ln x) + \int x (-\sin(\ln x)) (\ln x)' dx \\
&= x \sin(\ln x) - x \cos(\ln x) - \int \sin(\ln x) dx && \left| \begin{array}{l} \text{add } \int \sin(\ln x) dx \\ \text{to both sides} \end{array} \right. \\
2 \int \sin(\ln x) dx &= x \sin(\ln x) - x \cos(\ln x) \\
\int \sin(\ln x) dx &= \frac{x}{2} (\sin(\ln x) - \cos(\ln x)) \quad .
\end{aligned}$$

**Solution.** 7.s

$$\int \ln x dx = x \ln x - \int x d(\ln x) = x \ln x - \int \frac{x}{x} dx = x \ln x - x + C \quad .$$

**Solution.** 7.u

$$\begin{aligned}
\int \frac{\ln x}{\sqrt{x}} dx &= \int (\ln x) 2d(\sqrt{x}) && \left| \text{integrate by parts} \right. \\
&= (\ln x) 2\sqrt{x} - \int 2\sqrt{x} d(\ln x) \\
&= 2\sqrt{x} \ln x - 2 \int \frac{\sqrt{x}}{x} dx \\
&= 2\sqrt{x} \ln x - 2 \int x^{-\frac{1}{2}} dx \\
&= 2\sqrt{x} \ln x - 4\sqrt{x} + C \\
&= 2\sqrt{x} (\ln x - 2) + C \quad .
\end{aligned}$$

8. Compute  $\int x^n e^x dx$ , where  $n$  is a non-negative integer.

**Solution.** 8

$$\begin{aligned}
\int x^n e^x dx &= \int x^n de^x \\
&= x^n e^x - \int e^x dx^n \\
&= x^n e^x - n \int x^{n-1} e^x dx \\
&= x^n e^x - n \left( \int x^{n-1} de^x \right) \\
&= x^n e^x - n \left( x^{n-1} e^x - \int (n-1) x^{n-2} e^x dx \right) \\
&= x^n e^x - n x^{n-1} e^x + n(n-1) \int x^{n-2} e^x dx \\
&= \dots (\text{continue above process}) \dots \\
&= x^n e^x - n x^{n-1} e^x + n(n-1) x^{n-2} e^x - \dots \\
&\quad + (-1)^k n(n-1)(n-2) \dots (n-k+1) x^{n-k} e^x \\
&\quad + \dots + (-1)^n n! e^x + C \\
&= C + \sum_{k=0}^n (-1)^k \frac{n!}{(n-k)!} x^{n-k} e^x \quad .
\end{aligned}$$